Student Performance Trend Analysis

Problem statement

Compare subject scores across examinations, calculate improvements and declines, identify consistently weak subjects, and classify students using defined criteria.

Assumptions and rules

• Each student has a list of scores in the same subject order for three examinations.

• A subject is consistently weak if the score is below 40 in every examination.

• Overall classification uses the average of all scores: 75 or more DISTINCTION, 60 or more FIRST CLASS, 50 or more SECOND CLASS, otherwise NEEDS IMPROVEMENT.

Algorithm

1. Store subject names and student scores for three examinations.

2. For each student and subject, calculate change from first to last examination.

3. Label the trend as improved, declined, or unchanged.

4. Identify subjects below the weak threshold in every examination.

5. Calculate the student's average score across all subjects and examinations.

6. Assign a classification using the stated criteria.

7. Display subject trends, weak subjects, average, and classification.

Pseudocode

```
START\nINPUT subjects and student score lists\nFOR each student\n  FOR each subject\n    change = last_score - first_score\
```

Flowchart

```
[Start] -> [Read subjects and exam scores]\n   |\n[Select student and subject]\n   |\n[Compare first and last scores]\n   |
```

Python solution

The program uses beginner-level Python constructs from the assigned syllabus: variables, data types, operators, conditions, loops, strings, lists, tuples, and unpacking where appropriate.

```
subjects = ["Maths", "Physics", "Chemistry", "English"]
students = [
    ("Anu", [
        [35, 38, 42],
        [55, 60, 64],
        [48, 45, 39],
        [70, 75, 80]
    ]),
    ("Bala", [
        [80, 82, 85],
        [72, 70, 68],
        [65, 67, 70],
        [60, 64, 66]
    ])
]

weak_limit = 40

for student in students:
    name, marks_by_subject = student
    total = 0
    count = 0
    weak_subjects = []

    print("\nStudent:", name)
    for i in range(len(subjects)):
```

```
        scores = marks_by_subject[i]
        change = scores[2] - scores[0]

        if change > 0:
            trend = "Improved"
        elif change < 0:
            trend = "Declined"
        else:
            trend = "Unchanged"

        if scores[0] < weak_limit and scores[1] < weak_limit and scores[2] < weak_limit:
            weak_subjects.append(subjects[i])

        total = total + sum(scores)
        count = count + len(scores)
        print(subjects[i], "Scores:", scores,
              "Change:", change, trend)

    average = total / count
    if average >= 75:
        classification = "DISTINCTION"
    elif average >= 60:
        classification = "FIRST CLASS"
    elif average >= 50:
        classification = "SECOND CLASS"
    else:
        classification = "NEEDS IMPROVEMENT"

    print("Consistently weak subjects:", weak_subjects)
    print("Overall average:", round(average, 2))
    print("Classification:", classification)
```

Sample output

```
Student: Anu\nMaths Scores: [35, 38, 42] Change: 7 Improved\nPhysics Scores: [55, 60, 64] Change: 9 Improved\nChemistry Sco
```

Explanation / observations

The sample stores three marks per subject. Trend is based on the first and last examinations, while weakness requires all three scores to be below 40.

Conclusion

This solution models the stated case using a clear sequence of input, processing, decisions, repetition, and output. The assumptions and sample data can be changed to test other cases.